The Robert M. Buchan Department of Mining

Queen's University

MNTC 313 – Assignment 4 Version A

Kingston's Water Tower needs to be drained once a year, and the mayor has hired you as the main engineer to make a few calculations about what the status of the draining is. Assume that the tank can be approximated by a cylinder of radius $R = 25\ m$. The average velocity of water leaving through the pipe at the bottom of the tank is not constant and is a function of the height of the water level in the tank. You can assume that the liquid outlet velocity when draining the tank container can be calculated as:

$$V = 0.5\sqrt{2 * gh}$$

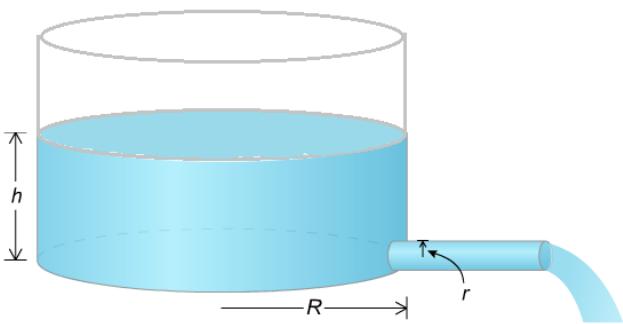
when r is much smaller than R, where

g = Acceleration due to gravity: 9.81 m/s2

V = outlet velocity (m/s)

h = height between surface and aperture (m)

Consider a value $r = 0.5\ m$.



Write a program in C/C++ that helps you calculate the following for the Water Tower with a given initial water heigh, h, cylinder of radius, R, and the radius of pipe at the bottom, r, considering time intervals of $t = 1\ seg$, in which you can assume a constant velocity:

- The heigh value of the water inside the tank after 1800 seconds (30 min).

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- The heigh value of the water inside the tank after *T* seconds (use a value of *T* less than 3600).
- The average value of the velocity when draining the tank after *T* seconds.
- The mayor wants to know the status of the draining after *T* seconds. Add a condition to output "Tank half drained" if the tank meets this criterion, or "Tank still not half drained" if it does not.

Hint:

1) use a for-loop statement ***for(int t = 0; t <= finalT; t++)***
2) compute the velocity of water leaving through the pipe at the bottom of the tank at iteration ***t***
3) update the first the volume in the tank, and then the current height based on the volume left. The volume that leaves the tank is given by the

   Vol [m^3]= Q [m^3/seg] * Δtime [seg] = *V* [m/seg]* *cross section area of pipe* [m^2]*1 [seg]

Your output must look like:

```
15,25,0.5,3600
Average velocity value: 6.81 m/s
Height value after 30 min: 9.46 m
Height value after 3600 seconds: 5.19 m
Tank half drained
```

Note: **Please refer to the AutoGradr test cases for the desired output format.** A starter code file has been provided on OnQ. Download the file, and copy and paste its contents into a new .c/.cpp file in your IDE. Round outputted values to the nearest whole number. Use pi = 3.14.

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